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Artificial Intelligence and Persuasion: A Construal-Level Account

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Abstract

Although more individuals are relying on information provided by nonhuman agents, such as artificial intelligence and robots, little research has examined how persuasion attempts made by nonhuman agents might differ from persuasion attempts made by human agents. Drawing on construal-level theory, we posited that individuals would perceive artificial agents at a low level of construal because of the agents' lack of autonomous goals and intentions, which directs individuals' focus toward *how* these agents implement actions to serve humans rather than *why* they do so. Across multiple studies (total $N = 1{,}668$), we showed that these construal-based differences affect compliance with persuasive messages made by artificial agents. These messages are more appropriate and effective when the message represents low-level as opposed to high-level construal features. These effects were moderated by the extent to which an artificial agent could independently learn from its environment, given that learning defies people's lay theories about artificial agents.

Keywords

artificial intelligence, robot, persuasion, construal-level theory, open data, open materials, preregistered

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Intelligent nonhuman agents, such as artificial intelligence and robots, referred to as *artificial agents* by researchers (Floridi & Sanders, 2004; Huang & Rust, 2018; Russell & Norvig, 2010), are increasingly common across a variety of platforms and applications created to interact with human decision makers. Artificial agents (AAs) have proven to be superior to humans in making financial investments and diagnosing certain diseases (e.g., skin cancer; Esteva et al., 2017). Despite these advances, research has shown many people are still averse to interactions with AAs (Dietvorst, Simmons, & Massey, 2015). In the current research, we drew on construal-level theory (Trope & Liberman, 2010) to examine how AA systems can increase persuasive effectiveness, a previously unexamined dimension within people's mental representations of AAs.

First, this research identified a key linkage between AAs and construal-level perceptions among humans with whom AAs interact. According to action-identification theory (Vallacher & Wegner, 1989), an action (e.g., "painting a room") can be represented at a high level of construal emphasizing why the action is conducted (e.g., "making the room look fresh") or at a low level of construal processing how the action is conducted (e.g., "applying brush strokes"). Construal-level theory posits that whether the actor is represented at a high level or a low level of construal depends on the presence of superordinate goals associated with the actor's action (Fujita, Trope, Liberman, & Levin-Sagi, 2006; Trope & Liberman, 2010). Notably, AAs are designed and programmed by humans to serve, and the common lay theory that individuals hold is that AA's actions are not driven by their own autonomous goals or intentions (Russell & Norvig, 2010; Ward, Olsen, & Wegner, 2013). Research has shown that this lack of perceived intention in agents inhibits the ability to infer superordinate goals derived from the agent, thus leading to low-level identifications for the actor among observers (Kozak, Marsh, & Wegner, 2006). Therefore, we hypothesized that individuals

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would interpret AAs' actions at lower level construals as a default (i.e., focusing on how rather than why) compared with the same action performed by humans. We posited that when AAs are perceived at lower-level construals, they would be perceived as possessing fewer superordinate goals than agents perceived at higher-level construals.

The persuasion literature has identified a matching effect whereby a fit among salient characteristics of the persuasion source, persuasion recipient, and the persuasion message increase the effectiveness of the message (Cesario, Grant, & Higgins, 2004; Teeny, Briñol, & Petty, 2017). For example, research has shown that a match occurs when the appropriate conjunction of source and message increases the extent to which recipients' experience "feeling right" about or validated by the persuasive appeal or source (Briñol & Petty, 2009; Cesario et al., 2004). We hypothesized that AAs are low-construal agents lacking high-construal abilities. Thus, we posited that individuals would perceive greater congruence and appropriateness when the persuasion message of AAs endorses low as opposed to high-construal attributes, leading to more effective persuasion.

Further, we hypothesized that informing individuals that AAs possess learning capabilities would attenuate low-construal beliefs about AAs for several reasons. First, learning involves abstraction and generalization from one's repeated experiences, which is a high-construal mental process (Trope & Liberman, 2010). Second, learning is associated with metacognitive awareness of the self (Ridley, Schutz, Glanz, & Weinstein, 1992) and is likely to attenuate the perception that the AA is simply a program designed by humans without understanding the meaning and purpose of its actions. Additionally, learning is associated with other high-construal constructs, such as goal generation, autonomy, intrinsic motivation, and deriving meanings from one's experiences (Dweck, 1999; Niiya, Crocker, & Bartmess, 2004; Ridley et al., 1992). Thus, we posited that individuals would perceive high-construal messages as less appropriate when coming from AAs.

Overview of Experimental Studies

In Studies 1a, 1b, and 1c, we provided initial evidence that AAs conducting low- as opposed to high-construal actions would be perceived as more appropriate. In Study 2, we showed that messages from AAs are more persuasive when they highlight low-level as opposed to high-level construals. In Studies 3, 4, and 5, we extended these effects to interactions using distinct types of AAs and further showed that changing an AA's perceived characteristics attenuates and even reverses low-construal perceptions of the AA.

Study 1a

In Study 1a, we examined whether individuals differ in their interpretations of AA's and humans' actions with respect to construal level.

Method

Participants and design. Study 1a had 100 participants from Amazon Mechanical Turk (MTurk; women = 41%, mean age = 35.57 years). The average effect size (i.e., $d = 0.63$) reported in previous AA research with similar experimental designs (Waytz, Heafner, & Epley, 2014; Waytz & Norton, 2014) was used to determine that at least 40 participants would be needed per experimental condition. This sample size provides sufficient power (> 80%) to detect the effect if it exists (Bausell & Li, 2002). Additionally, sample size was influenced by the size of the participant pool made available to the authors in the given semester. In Study 1a, each participant was randomly assigned to one of the conditions in a single-factor (agent: human, AA) between-participants design.

Procedure. The key dependent variable was the Behavioral Identification Form (BIF), a scale developed to measure the extent to which an individual perceives an action to be consistent with either low- or high-level construals (Vallacher & Wegner, 1989). Among the 25 BIF items, 7 items (Items 3, 13, 15, 16, 19, 20, 23) were not used because they did not make sense as actions to be conducted by an AA (e.g., "eating"; see the section on Study 1a in the Supplemental Material available online for more information).

First, participants were instructed to imagine that the actions provided in the BIF scale were conducted by either a human or an AA, depending on condition. Then, participants chose the descriptions that they thought were more appropriate for the agent that they were assigned to (i.e., either a human or an AA). It was our prediction that individuals who imagined that the actions in the BIF scale were conducted by an AA would choose a higher proportion of low-construal descriptions for the given actions. Finally, participants answered demographic questions and were debriefed.

Results

We summed the number of high-construal choices made by each participant to create an index of construal level. The index ranged from 0 to 18. The index of construal level was submitted to a single-factor (agent: human, AA)

one-way analysis of variance (ANOVA). We found that the construal-level score was significantly lower when participants imagined that the action was conducted by an AA ($M = 4.04$, $SD = 5.51$, 95% confidence interval, or CI = [2.44, 5.64]) compared with when the action was conducted by a human ($M = 11.28$, $SD = 5.88$), 95% CI = [9.68, 12.88]), $F(1, 98) = 40.36$, $p < .001$, $\eta_p^2 = .29$.[1]

Study 1b

Study 1b was designed to provide further evidence that actions performed by AAs are perceived as more appropriate when they are described in low-construal as opposed to high-construal terms.

Method

Participants and design. Study 1b had 191 undergraduate business students (women = 49%, mean age = 20.88 years). Each participant was randomly assigned to one of the conditions in a 2 (agent: human, AA) × 2 (action description: high construal, low construal) between-participants design.

Procedure. The procedure was similar to that used in Study 1a. Participants were provided with 17 sentences adapted from the BIF and revised to describe the same action in either high-construal or low-construal terms (Vallacher & Wegner, 1989). An example of a high-construal statement was, "[Agent] is locking a door to secure the house." An example of a low-construal statement was, "[Agent] is locking a door by putting a key in the lock." In the AA condition, the agent was "a robot." In the human condition, the agent was "a person" (see the section on Study 1b in the Supplemental Material for more information).

Then, we measured the statements' appropriateness using three items ("appropriate," "natural," "makes sense"; 1 = *not at all*, 7 = *very much*); the three items were averaged to create an appropriateness score ($\alpha = .86$). Also, the appropriateness scores for all 17 actions were averaged to create an index of appropriateness ($\alpha = .92$). Finally, participants answered demographic questions and were debriefed.

Results

The index of appropriateness was submitted to a 2 (agent: human, AA) × 2 (action description: high construal, low construal) ANOVA. The main effect of agent on appropriateness was significant: The human conditions reported higher appropriateness ($M = 4.83$, $SD = 0.96$) than the AA conditions ($M = 4.35$, $SD = 1.22$), $F(1, 187) = 9.21$, $p = .003$, $\eta_p^2 = .05$. The main effect was qualified by a two-way interaction between agent and action description, $F(1, 187) = 6.97$, $p = .009$, $\eta_p^2 = .04$. Additional pairwise comparisons across the AA conditions showed that low-construal descriptions ($M = 4.59$, $SD = 1.21$, 95% CI = [4.23, 4.95]) were perceived as more appropriate than high-construal descriptions ($M = 4.11$, $SD = 1.19$, 95% CI = [3.77, 4.46]), $F(1, 187) = 4.62$, $p = .033$, $\eta_p^2 = .04$. In the human condition, the appropriateness index was not significantly different in the high-construal description ($M = 5.00$, $SD = 0.92$, 95% CI = [4.73, 5.26]) and the low-construal description conditions ($M = 4.65$, $SD = 0.98$, 95% CI = [4.37, 4.94]), $F(1, 187) = 2.49$, $p = .12$, $\eta_p^2 = .03$. Additionally, the high-construal description was perceived to be more appropriate when the agent was a human ($M = 5.00$, $SD = 0.92$, 95% CI = [4.73, 5.26]) than when the agent was an AA ($M = 4.11$, $SD = 1.19$, 95% CI = [3.77, 4.46]), $F(1, 187) = 16.36$, $p < .001$, $\eta_p^2 = .15$ (see Fig. 1). Thus, this study provides further evidence that participants think of AAs' actions in terms of low construal.

Study 1c: Effect of AA Learning on Construal Level and Superordinate Goals

Study 1c examined whether the effect of AAs on construal level extends to a persuasion context and whether recommendations given by AAs differ with respect to construal level using a different measure of construal based on desirability (i.e., high construal) and feasibility (i.e., low construal) perceptions (Liberman & Trope, 1998). Study 1c also compared AAs as a function of their learning capabilities. We examined the moderating role of learning capabilities to assess whether beliefs about learning impact perceptions of the goals of AAs; we predicted that higher-construal agents would be perceived to have more superordinate goals. In Study 1c, we also measured alternative source characteristic factors other than construal level to determine their potential impact.

Method

Participants and design. Each of 119 undergraduate students (women = 44%, mean age = 21.39 years) was randomly assigned to one of the conditions in a single-factor (agent: learning AA, nonlearning AA) between-participants design.

Procedure. The experiment took place in the Informatics Department's robotics lab in which the industrial robot Baxter (ReThink Robotics, Rheinböllen, Germany) was located. The study was conducted on participants' cell phones because computer monitors were not available in

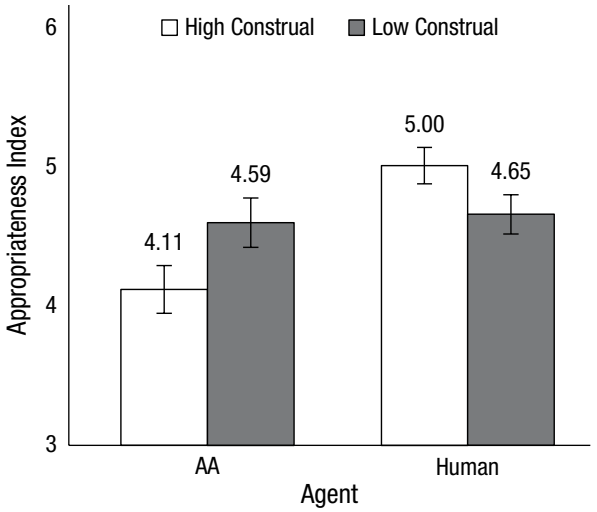


Fig. 1. Mean score on the appropriateness index as a function of agent type and construal level (Study 1b). Error bars indicate standard errors. AA = artificial agent.

the robotics department. In particular, we sent the link for the study when participants arrived, and they opened the link on their cell phone and participated in the experiment with their phones. Individuals participated in the study in groups of 3, and they interacted with Baxter across a table. The participants were told in a cover story that Baxter would make a unique product recommendation on the basis of their personal information.

Baxter welcomed the participants verbally and also by waving its hand. Then participants in the learning condition were informed that Baxter had a learning capability and could learn and improve its capabilities from experience; participants in the nonlearning condition were informed that Baxter did not have a learning capability and could not learn and improve its capabilities from experience. Then participants filled out a questionnaire purported to measure their risk of skin cancer (e.g., age, gender, residential area's average temperature, hours of daily sunlight exposure, history of skin cancer in the family). While participants were waiting for Baxter to create a personalized product recommendation based on the information they provided in the questionnaire, they were asked to list up to 10 goals that Baxter may have. Previous research suggests that having independent judges analyze a participant's writing is an effective way to determine the extent to which the writer is actively thinking about certain constructs. For example, the linguistic-categorization model (Semin & Fiedler, 1988) suggests that words used in a participant's writing can be analyzed by judges and categorized into four linguistic groups defined along a dimension of concreteness to abstractness, and then the overall concreteness of the writing can be evaluated. Thus, we recruited and trained two independent judges who analyzed the participants' writing and rated the extent to which participants inferred superordinate goals (i.e., capabilities to generate one's own goals and act to achieve them; Bagozzi & Dholakia, 1999) about Baxter's action ("To what extent does the writing reflect the writer's belief that Baxter has superordinate goals?" 1 = *not at all*, 5 = *very much*; the two judges' ratings were averaged to create the index of superordinate-goal inference, $\alpha = .88$; for more information about the judges' rating instructions and training procedure, see the section on Study 1c in the Supplemental Material).

After participants completed their writing on Baxter's potential goals, they received a message from Baxter recommending the use of spray-type sunscreen. To examine whether perceptions of Baxter's learning capabilities influenced construal level, we instructed participants to write down potential benefits of the recommended product. Research has shown that individuals who focus on high as opposed to low construal think more about an object's desirability as opposed to feasibility features (Liberman & Trope, 1998). Because we hypothesized that nonlearning AAs are low-construal agents, endowing them with learning capabilities would attenuate our construal effect. Thus, it was our prediction that a learning Baxter would lead participants to generate more desirability-related as opposed to feasibility-related thoughts compared with a Baxter without the ability to learn from its experiences. Following the same procedure as in the superordinate-goal analysis, two judges analyzed the product-benefits writing and rated the extent to which participants focused more on sunscreen's attributes that were related to desirability as opposed to feasibility (1 = *not at all*, 5 = *very much*; two judges' ratings were averaged to create the index of desirability focus; for more information about the judges' rating instructions and training procedures, see the section on Study 1c in the Supplemental Material).

We also included several additional source-characteristic measures that may affect perceptions of AAs beyond construal level. One measure assessed thoughts about the AA's creator ("When you were interacting with Baxter, to what extent did you think about the person or the group of people who created Baxter?" 1 = *not at all*, 7 = *very much*). Also, we measured the level of "trust," "likability," and "expertise" participants felt about Baxter as a source of recommendations (1 = *not at all*, 7 = *very much*). Participants answered a manipulation-check question designed to assess our learning-capabilities manipulation ("Baxter can only conduct the task that it is programmed to do"; 1 = *strongly disagree*, 7 = *strongly agree*, reverse-coded to

represent the perception of learning capabilities). Finally, participants answered demographic questions and were debriefed.

Results

First, an ANOVA revealed that participants in the nonlearning condition had stronger beliefs that "Baxter can only conduct the task that it is programmed to do" (M = 4.93, SD = 1.57, 95% CI = [4.53, 5.34]) compared with participants in the learning condition (M = 4.29, SD = 1.60, 95% CI = [3.87, 4.70]), $F(1, 117) = 4.93$, $p = .028$, $\eta_p^2 = .04$. This indicates that the manipulation of the AA's learning capability was successful.

An ANOVA on the index of superordinate-goal inference (i.e., judges' evaluations of "To what extent does the writer believe that Baxter has superordinate goals?" 1 = *not at all*, 5 = *very much*) showed that participants inferred greater superordinate goals from Baxter in the learning condition (M = 3.14, SD = 0.84, 95% CI = [2.92, 3.35]) than in the nonlearning condition (M = 2.68, SD = 0.78, 95% CI = [2.48, 2.89]), $F(1, 117) = 9.26$, $p = .003$, $\eta_p^2 = .07$. This finding is consistent with the idea that lower-construal agents are perceived to possess fewer superordinate goals.

An ANOVA on the index of relative desirability-feasibility focus of product benefits showed that participants generated more desirability-related benefits in the learning-Baxter condition (M = 3.17, SD = 1.36, 95% CI = [2.81, 3.53]) than in the nonlearning-Baxter condition (M = 2.64, SD = 1.32, 95% CI = [2.30, 2.98]), $F(1, 117) = 4.59$, $p = .034$, $\eta_p^2 = .04$. Also, there was no difference in writing length measured by the number of total words used in writing, $p = .44$.

Finally, we analyzed the additional source-characteristic measures of creator-related thoughts, perceived trust, likability, and expertise attributed to Baxter. We found that participants' thoughts related to the creator of Baxter were not significantly different between the learning condition (M = 3.63, SD = 1.72, 95% CI = [3.18, 4.08]) and the nonlearning condition (M = 3.87, SD = 1.68, 95% CI = [3.43, 4.30]), $F(1, 117) = 0.59$, $p = .44$, $\eta_p^2 = .005$. Similarly, there were no differences between the learning and nonlearning Baxter regarding perceived trust, $p = .20$; likability, $p = .57$; and expertise, $p = .67$, thus indicating that learning and nonlearning AAs did not differ in other persuasion-relevant source characteristics.

Discussion

Study 1c found additional evidence of the link among AAs, perceptions of learning capabilities, and construal using different measures. Study 1c found that perceptions of AAs as low-construal agents were attenuated when AAs were, as opposed to were not, capable of learning, because learning perceptions led to higher superordinate-goal beliefs. However, perceptions of learning capabilities did not increase thoughts about the AA's creator or the trust, likability, and expertise of the source, thus indicating that these sources were perceived similarly on key dimensions, except for construal level.

Study 2

The purpose of Study 2 was to test whether a message from an AA is more persuasive when the message is framed in low-level as opposed to high-level construal terms.

Method

Participants and design. Three hundred twenty-four participants from MTurk (women = 56%, mean age = 37.30 years) were each randomly assigned to one of the conditions in a 2 (agent: human, AA) × 2 (message: high construal, low construal) between-participants design.

Procedure. Participants in the human conditions were instructed to browse an online medical website (www.sermo.com) and were told that they would receive instant medical advice from a doctor online. Participants in the AA conditions were instructed to browse a web page describing the abilities of IBM's artificial intelligence robot Watson's to medical diagnosis of skin cancer (https://www.ibm.com/cognitive/au-en/melanoma) and were told that they would receive instant medical advice from Watson. In a posttest with 120 MTurk participants, we measured the extent to which participants trusted, liked, and perceived expertise in the agents as a recommendation source (1 = *not at all*, 7 = *very much*). The results of this analysis confirmed that individuals did not perceive the AA and the human doctor to differ significantly in levels of trust (AA: M = 4.79; human: M = 5.07), expertise (AA: M = 5.10; human: M = 5.29), and likability (AA: M = 4.87; human: M = 5.09), all ps > .23. The posttest also measured participants' construal level using the full 25-item BIF scale. In this case, we used a full BIF scale to measure the current construal level of the participants after visiting the website that portrayed a human agent or an AA Watson. Unlike in the previous studies, in which we used a reduced BIF scale because some items were not appropriate for AAs, we used a full BIF scale in Study 2 because the objective was to examine whether participants' construal level was differentially affected by interaction with a human as opposed to an AA. As predicted, participants in the AA-doctor condition reported a lower level of construal compared with the human-doctor

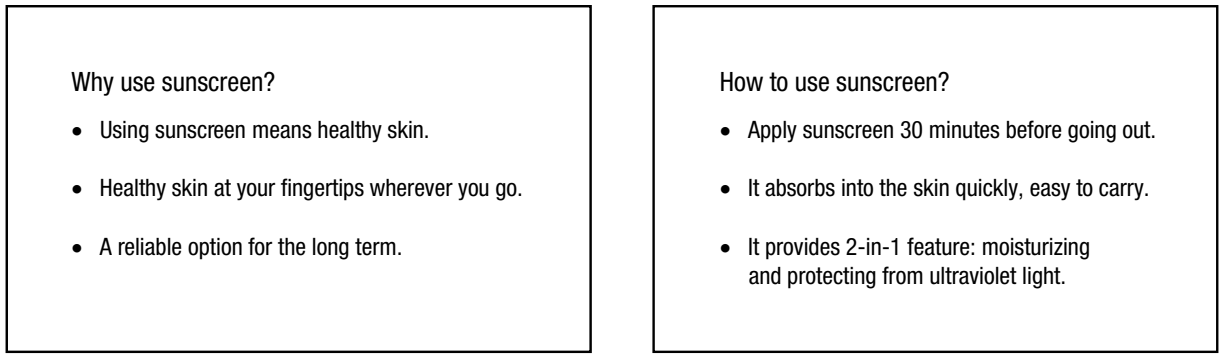


Fig. 2. Sunscreen recommendations made to participants using a (left) high-construal message and (right) low-construal message (Study 2).

condition (AA BIF score: $M = 14.73$; human BIF score: $M = 17.22$), $F(1, 118) = 5.94$, $p = .016$, $\eta_p^2 = .05$. Therefore, the two agents influenced construal level differently, but they were not perceived differently with respect to other key source characteristics.

In Study 2, after browsing either the human doctor's or IBM Watson's medical website, participants filled out a questionnaire that purported to measure their risk of skin cancer (e.g., age, gender, residential area, hours of daily sunlight exposure, history of skin cancer in the family). Then they received a medical advice message from either Watson or a human doctor based on their answers. On the basis of prior construal-level research (Han, Duhachek, & Agrawal, 2016; Lee, Keller, & Sternthal, 2010), we created the message to highlight either high-level or low-level construal prescriptive statements regarding sunscreen use (see Fig. 2).

Next, participants indicated their intention to apply sunscreen using measures adapted from Keller (2006; e.g., "I intend to use sunscreen"; 1 = *very unlikely*, 7 = *very likely*); the four items were averaged to create an index of intention, $\alpha = .91$. Then participants answered demographic questions and were debriefed.

Results

We first submitted the intention to use sunscreen to a 2 (agent: human, AA) × 2 (message: high construal, low construal) ANOVA. The result revealed that the main effects of agent, $F(1, 320) = 0.94$, $p = .333$, $\eta_p^2 = .003$, and message on the intention to use sunscreen, $F(1, 320) = 2.27$, $p = .13$, $\eta_p^2 = .007$, were both nonsignificant. The hypothesized interaction between agent and message was significant, $F(1, 320) = 7.26$, $p = .007$, $\eta_p^2 = .02$. Additional pairwise comparisons revealed that when the message was coming from an AA, the low-construal message ($M = 5.05$, $SD = 1.48$, 95% CI = [4.72, 5.37]) was more effective than the high-construal message ($M = 4.31$, $SD = 1.65$, 95% CI = [3.94, 4.67]) in persuading individuals to use sunscreen, $F(1, 320) = 8.82$, $p = .003$, $\eta_p^2 = .05$. Also, the low-construal message led to a higher intention to use sunscreen when it was coming from an AA ($M = 5.05$, $SD = 1.48$, 95% CI = [4.72, 5.37]) than when the message was coming from a human ($M = 4.40$, $SD = 1.60$, 95% CI = [4.04, 4.76]), $F(1, 320) = 6.75$, $p = .010$, $\eta_p^2 = .04$ (see Fig. 3). Additionally, when the message was coming from a human, there was no significant difference between the low-construal message ($M = 4.40$, $SD = 1.60$, 95% CI = [4.04, 4.76]) and the high-construal message ($M = 4.61$, $SD = 1.62$, 95% CI = [4.25, 4.97]) in persuading individuals to use sunscreen, $F(1, 320) = 0.71$, $p = .40$, $\eta_p^2 < .01$.

Discussion

Study 2 showed that low-construal as opposed to high-construal messages were more effective when the message was delivered by an AA but not when the source was human. This finding extends the results of our previous studies into a persuasion context and demonstrates that persuasion from AAs is enhanced when coupled with a low-construal message.

Study 3

Study 3 had several goals. First, we sought to replicate the persuasion effect using a different AA (i.e., Amazon's Alexa). Second, we examined whether endowing an AA with learning capabilities would attenuate the persuasion effect. Finally, we sought additional evidence that persuasion occurs via our hypothesized superordinate-goal perception mechanism. Study 3 also measured additional source characteristics to ensure that they were not viable alternatives to construal.

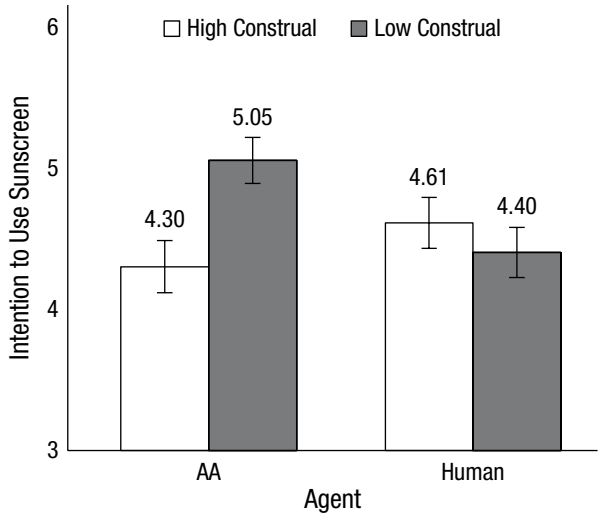


Fig. 3. Mean intention to use sunscreen as a function of agent type and message's construal level (Study 2). Error bars indicate standard errors. AA = artificial agent.

Method

Participants and design. Two hundred and thirty-four undergraduate business students (women = 54%, mean age = 20.09 years) were each randomly assigned to one of the conditions in a 2 (agent: learning AA, nonlearning AA) × 2 (message: high construal, low construal) between-participants design.

Procedure. The study took place in the university behavioral lab. After arrival to the lab, participants were seated separately in 19 cubicles in front of a computer. In the middle of the lab space, Echo, an AA created by Amazon, was placed on a table where it was visible to each participant. Participants were given a cover story that they would receive a personalized product recommendation from Amazon Echo's virtual agent Alexa.

To manipulate beliefs about Alexa's capability, we randomly assigned each participant to one of two conditions in which Alexa was described as an agent with either a "fixed intelligence" or "experiential learning capabilities." Visual stimuli were also provided (see Figs. 4 and 5). Then Alexa asked participants to list up to 10 items that they had purchased recently. Next, Alexa told participants that she had received their past purchase histories and that she would send personalized product recommendations to each participant's computer screen. While participants were waiting for their recommendation, they answered a measure of superordinate-goal perception ("To what extent to do you think Alexa can generate its own goals and act to achieve them?" 1 = *not at all*, 7 = *very much*).

Then, on their computers, each participant randomly received either a high-construal message or a low-construal message that advertised a gym. This stimulus was adapted from the work of Han et al. (2016) but was modified for our purposes (see Fig. 6). A pretest with 80 MTurk participants confirmed that participants who were exposed to the two ads showed comparable intentions to visit the gym (why ad: $M = 2.65$; how ad: $M = 2.98$), $F(1, 78) = 0.73$, $p = .396$.

Then participants indicated the extent to which they intended to visit the gym ("How likely is it that you would try out this gym in case you were looking for a place at which to work out?" 1 = *not at all likely*, 7 = *very likely*), which served as the measure of persuasive effectiveness. To measure whether the manipulation was successful, we also measured the extent to which participants believed that AAs cannot learn ("Alexa can only conduct the task that it is programmed to do"; 1 = *strongly disagree*, 7 = *strongly agree*). This measurement was reverse-coded to represent the belief that the AA can learn. Also, we measured the level of likability, trust, and expertise that participants felt about Alexa as a source of recommendation (1 = *not at all*, 7 = *very much*). Finally, participants answered demographic questions and were debriefed.

Results

First, belief in the AA's learning capability was submitted to a 2 (agent: learning AA, nonlearning AA) × 2 (message: high construal, low construal) ANOVA to assess the effectiveness of this manipulation. The results revealed that only the main effect of the manipulation of the AA's capability was significant; specifically, participants in the learning condition had a stronger belief that Alexa can learn ($M = 3.71$, $SD = 1.52$, 95% CI = [3.43, 3.98]) than did participants in the no-learning condition ($M = 3.10$, $SD = 1.53$, 95% CI = [2.82, 3.38]), $F(1, 230) = 9.50$, $p = .002$, $\eta_p^2 = .04$. Therefore, our manipulation of the AA's learning capability was successful. The level of likability, trust, and expertise did not differ depending on the AA's learning capability (all ps > .15) or message manipulation (all ps > .10; see the section on Study 3 in the Supplemental Material for more information).

The intention to visit the gym was submitted to a 2 (agent: learning AA, nonlearning AA) × 2 (message: high construal, low construal) ANOVA. The main effect of message was significant, $F(1, 230) = 25.03$, $p < .001$; specifically, the low-construal message ($M = 4.56$, $SD = 2.01$) was more persuasive than the high-construal message ($M = 3.28$, $SD = 1.94$). This main effect was qualified by a significant two-way interaction between agent

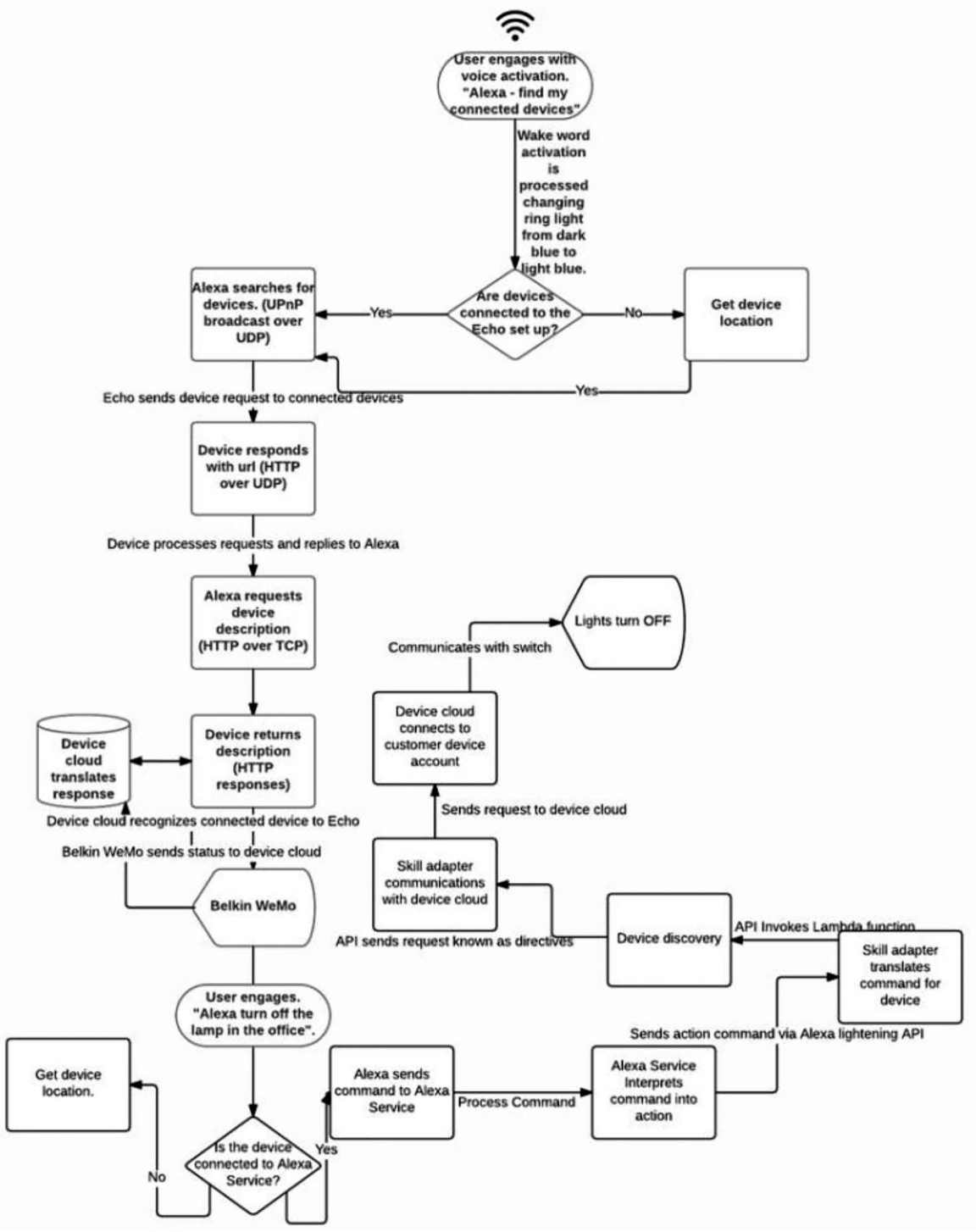


Fig. 4. Stimulus for the condition in which Alexa was framed as a fixed-capabilities agent (Study 3).

and message, $F(1, 230) = 12.00$, $p = .001$, $\eta_p^2 = .05$. Additional pairwise comparisons in the nonlearning-AA condition showed that the low-construal message ($M = 4.98$, $SD = 1.59$, 95% CI = [4.58, 5.39]) was more persuasive than the high-construal message ($M = 2.84$, $SD = 1.57$, 95% CI = [2.43, 3.26]), $F(1, 230) = 36.15$, $p < .001$, $\eta_p^2 = .32$. In the learning-AA condition, the high-construal message ($M = 3.70$, $SD = 2.17$, 95% CI = [3.14, 4.26]) and the low-construal message ($M = 4.09$, $SD = 2.31$, 95% CI = [3.47, 4.71]) were not significantly different in persuasiveness, $F(1, 230) = 1.17$, $p = .28$, $\eta_p^2 = .008$ (see Fig. 7). Additionally, the high-construal

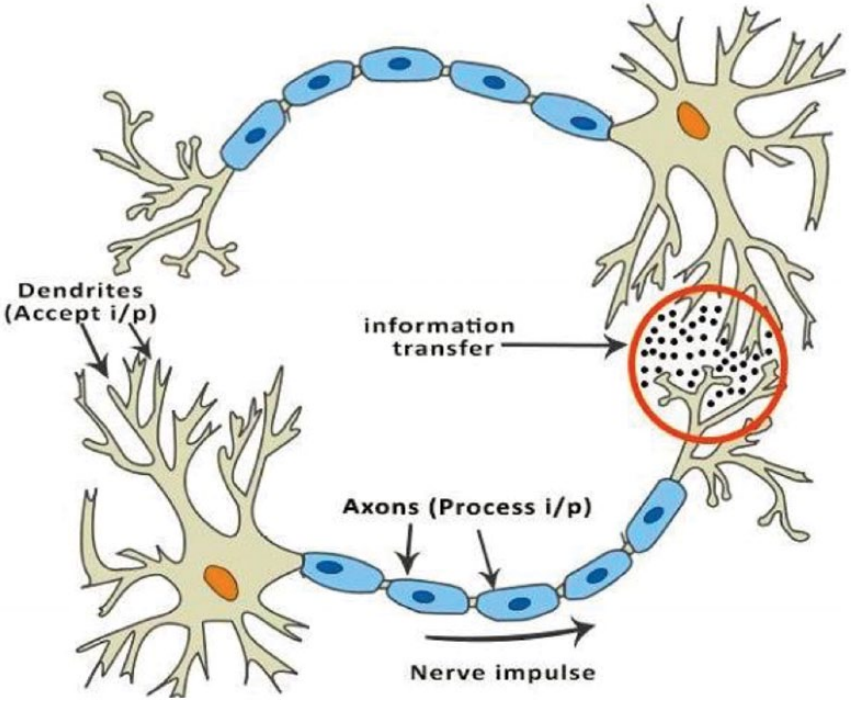


Fig. 5. Stimulus for the condition in which Alexa was framed as a learning-capabilities agent (Study 3).

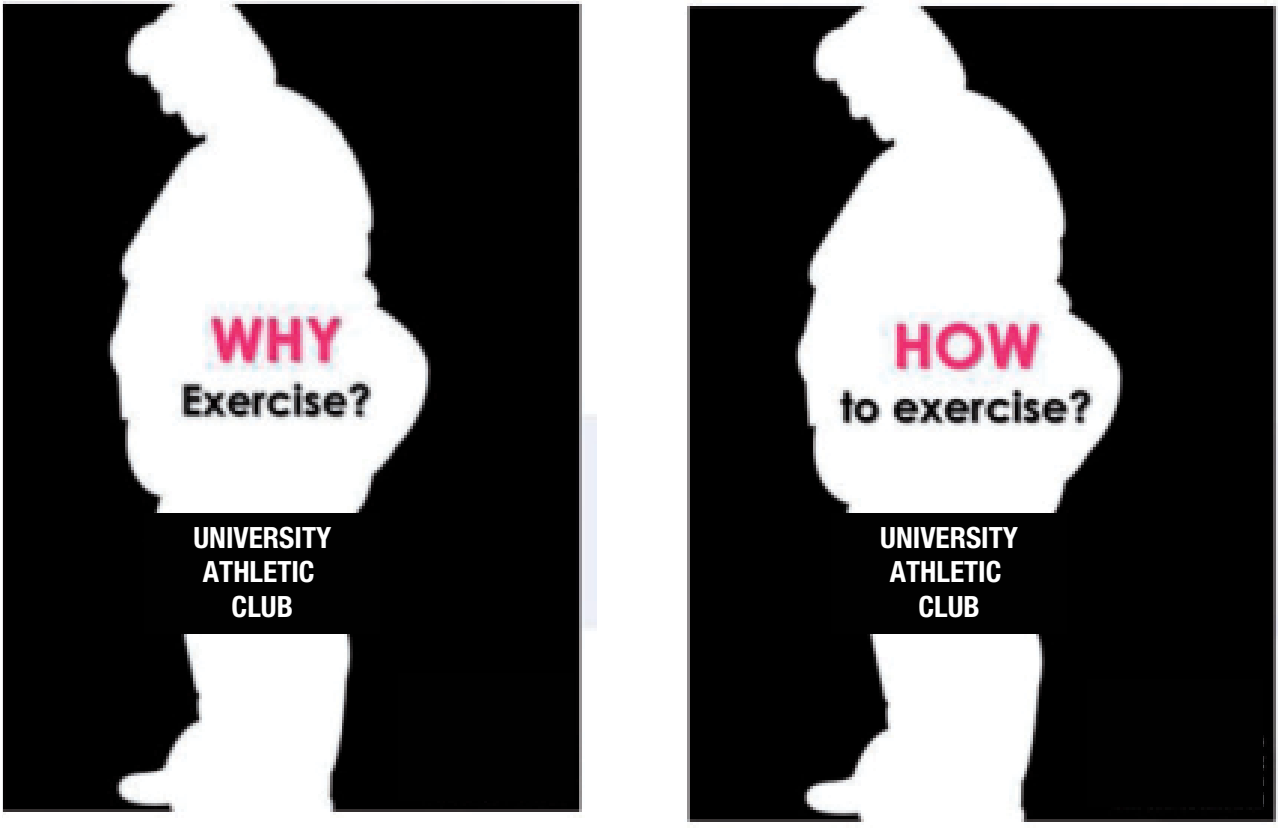


Fig. 6. Gym recommendation made to participants using a (left) high-construal message and (right) low-construal message (Study 3).

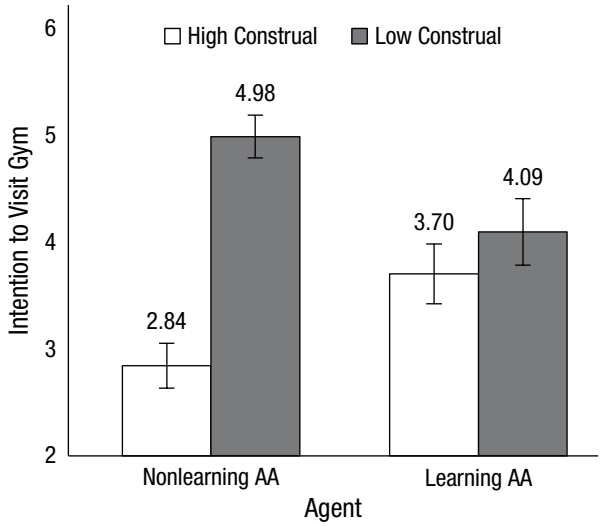


Fig. 7. Mean intention to visit the gym as a function of whether participants were told that the artificial agent (AA) could learn and message's construal level (Study 3). Error bars indicate standard errors.

message was more persuasive in the learning condition (M = 3.70, SD = 2.17, 95% CI = [3.14, 4.26]) than in the nonlearning-AA condition (M = 2.84, SD = 1.57, 95% CI = [2.43, 3.26]), $F(1, 230) = 5.75$, $p = .017$, $\eta_p^2 = .05$. In contrast, the low-construal message was more persuasive in the nonlearning-AA condition (M = 4.98, SD = 1.59, 95% CI = [4.58, 5.39]) than in the learning-AA condition (M = 4.09, SD = 2.31, 95% CI = [3.47, 4.71]), $F(1, 230) = 6.25$, $p = .013$, $\eta_p^2 = .05$.

To examine whether perceptions of superordinate goals mediated the effect of the key theorized interaction on persuasion, we conducted a mediation analysis. In particular, we followed the procedure outlined by Hayes and Preacher (2014) and compared the relative indirect effect for all four experimental conditions. According to Hayes and Preacher, "Evidence that at least one relative indirect effect is different from zero supports the conclusion that M mediates the effect of X on Y" (p. 462). In our case, the mediating variable (M) was the perception of superordinate goals, and the dependent variable (Y) was the intention to visit the gym. The independent variable (X) had four different combinations of conditions: low-construal message combined with a nonlearning AA (Condition 1), low-construal message combined with a learning AA (Condition 2), high-construal message combined with a nonlearning AA (Condition 3), and high-construal message combined with a learning AA (Condition 4). Therefore, this analysis produced four different indirect effects. Next, we compared each possible pair of indirect effects to examine whether any indirect effect was significantly different from another indirect effect. This analysis revealed that the indirect effect from Condition 1 was significantly different from Condition 4 (95% CI = [–0.2968, –0.0107]). This result implies that the effect of the independent variables on behavioral intentions was mediated by perceptions of superordinate goals (Hayes & Preacher, 2014).

Discussion

The results of Study 3 replicated the matching effect between a nonlearning AA and low-construal messages on persuasion. Study 3 also found evidence that this persuasion effect occurs because of changes in beliefs about the superordinate goals that AAs possess. When AAs are given the ability to learn, the low-construal matching effect on persuasion is attenuated.

Study 4

In Study 4, we sought to replicate the persuasion findings using a different type of AA, namely the humanoid robot Baxter. We also examined the role of learning capabilities using a measured variable capturing individual differences in beliefs in Baxter's learning capabilities.

Method

Participants and design. Each of 100 undergraduate business students (women = 57%, mean age = 20.50 years) was randomly assigned to one of the conditions in a single-factor (message: high construal, low construal) between-participants design.

Procedure. The experiment took place in the Informatics Department's robotics lab. As in Study 1c, Baxter asked participants to list up to 10 items they had purchased recently. Then Baxter told participants that it had received their past purchase histories and that it would create personalized product recommendations for each of them.

Then, just as in Study 3, participants randomly received either a high-construal message or low-construal message advertising a gym. Participants then indicated the extent to which they intended to visit the gym ("How likely is it that you would try out this gym in case you were looking for a place at which to work out?" 1 = *not at all likely*, 7 = *very likely*).

Also, as a manipulation check, we measured the extent to which participants believed Baxter cannot learn ("Baxter can only conduct the task that it is programmed to do"; 1 = *strongly disagree*, 7 = *strongly agree; M* = 5.05, *SD* = 1.46, 95% CI = [4.62, 5.48]). This measurement was reverse-coded to represent the belief that Baxter can learn. Finally, participants answered demographic questions and were debriefed.

Results

Eight participants did not follow instructions, communicating with others in their session and sharing their messages from Baxter. These participants were dismissed from the study, leaving 92 participants in the analysis. To assess persuasive effectiveness, we first submitted intention to visit the gym to a single-factor (message: high construal, low construal) ANOVA. The results revealed that the intention to visit the gym was greater in the low-construal message condition (M = 3.74, SD = 1.72, 95% CI = [3.25, 4.23]) than in the high-construal message condition (M = 2.93, SD = 1.60, 95% CI = [2.45, 3.42]), $F(1, 90) = 5.41$, $p = .022$, $\eta_p^2 = .06$.

To examine whether the message's construal level interacted with the measure of the AA's learning-capability belief, we conducted a regression analysis with the message-construal variable (high-construal message = 0, low-construal message = 1), the continuous measure of learning-capability belief, and their interaction predicting the intention to visit the gym. This regression analysis revealed a significant interaction between learning-capability beliefs and message construal, $b = -0.52$, $SE = 0.24$, 95% CI = [–0.05, –0.98], $t(88) = -2.19$, $p = .031$. To better understand this interaction, we used the Johnson-Neyman technique (Spiller, Fitzsimons, Lynch, & McClelland, 2013) to identify the range of learning-capability beliefs for which the simple effect of message construal was significant. This analysis revealed that low-construal as opposed to high-construal messages led to stronger behavioral intentions for individuals; the score for learning-capability beliefs was less than 3.23, b = 0.68, SE = 0.34, 95% CI = [0.00, 1.36], $p = .05$ (see Fig. 8). Thus, the low-construal message was more effective only for the individuals who held the belief that Baxter cannot learn (i.e., lower than 3.23 on a 7-point scale). Put differently, the relative effectiveness of low-construal as opposed to high-construal messages was attenuated for individuals who had a relatively stronger belief that Baxter can learn (i.e., higher than 3.23 on a 7-point scale). This effect replicated the findings of Study 3 with a measure of learning capabilities as opposed to the manipulation approach used in Study 3.

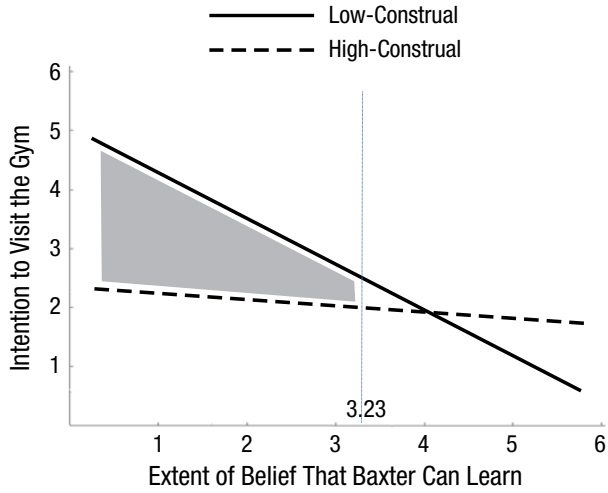


Fig. 8. Mean intention to visit the gym as a function of the extent to which participants believed that Baxter can learn and message's construal level (Study 4). Values below 3.23 on the y-axis indicate that the low-construal message was more effective than the high-construal message. Thus, the gray area represents the group with relatively lower learning belief and who were persuaded more by the low-construal message than by the high-construal message.

Study 5

Our previous studies showed that learning moderates people's perception of AAs. The objective of Study 5 was to examine whether manipulating other dimensions of AAs (e.g., consciousness) could reverse the match between agent and construal. Also, we examined whether depriving a human of certain features would liken the human to an AA, thereby creating a low-construal-message match for human agents. Also, we examined whether these effects were mediated by perceived fit between the agent and the message, providing additional evidence of the persuasion process.

Method

Participants and design. Four hundred MTurk participants (women = 52%, mean age = 35.64 years) were each randomly assigned to one of the conditions in a 2 (agent: human, AA) × 2 (typicality: typical, atypical) × 2 (message: high construal, low construal) between-participants design. The typicality manipulation refers to whether the agent was referred to as in previous studies (the typical case) or described as either a consciously self-aware AA or a dehumanized human agent (atypical case).

Procedure. Study 5 was introduced as a study that examines consumer preference and decision making. Just as in previous studies, participants listed up to 10 items that they had purchased recently and were told that a recommendation would be made to them on the basis of the information they provided. Participants were told that they would receive a recommendation from a recommendation specialist. The agent-manipulation stimuli were adopted from the work of Mende, Scott, van Doorn, Grewal, and Shanks (2019), and the description of each agent was revised to fit the purposes of Study 5. Participants were told that the recommendation message would be generated and delivered by either a human agent or an AA. Additionally, these agents' characteristics were described in either typical or atypical terms. Thus, there

were four agent conditions: typical human, atypical human (i.e., a robotlike human), typical AA, atypical AA (i.e., a humanlike AA). In the two human conditions, participants were shown a picture of a person adapted from the work of Mende et al. along with a brief introduction of the person, and they were told that the person would review the list of their recent purchase histories and make a recommendation. Participants in the atypical-human condition received additional information about the agent that was designed to make participants believe that the agent conducts tasks on the basis of algorithmically grounded decision rules (see Fig. 9). Previous research on technological dehumanization has shown that framing a person as lacking unique human features (e.g., depriving a person of intuition and spontaneity and instead highlighting regular and rule-based decision making) leads the person to be perceived as a mechanical dehumanized agent (Montague & Matson, 1983; Turkle, 1984, see Haslam, 2006, for a review). Thus, our manipulation was designed to liken the human agent to an AA that operates on the basis of algorithms.

In both the typical-AA and atypical-AA conditions, participants were shown a picture of a robot adapted from the work of Mende et al. (2019). However, the typical and the atypical AA were described differently to manipulate the degree to which the AA possessed humanlike qualities. In the typical-AA condition, participants were told that XT-1000, "an artificial intelligence robot," would review the list of their recent purchase history and make a recommendation. In the atypical-AA condition, the same agent was introduced with humanlike features. The agent was introduced with the name "Alex" instead of XT-1000. Mende et al. showed that manipulating features of the agent, such as varying names or other characteristics, could effectively alter individuals' beliefs about the agent. To further strengthen the difference between the atypical and the typical AA, we told participants that the AA possessed the human trait of consciousness (see Fig. 9 for more information). Among many features unique to humans, we chose consciousness, because previous research on an AA's consciousness showed that a self-conscious AA is a future goal of AA research and will enable the AA to learn, conduct abstract thinking, and generate its own goals (Dehaene, Lau, & Kouider, 2017). Extant research has also shown that envisioning consciousness among AAs is an ability most associated with viewing AAs as possessing equal or even superior intelligence to humans (Franklin, Wolpert, McKay, & Christian, 1997). Because of the strong conceptual association between the AA's consciousness and its high-construal capabilities, we predicted that framing an AA as being conscious as opposed to not conscious would make the AA be perceived as a high-construal agent.

Next, participants were each randomly assigned to one of the two message conditions. Just as in our previous studies, high-construal messages highlighted *why*, and the low-construal messages highlighted *how*. However, an important difference between the messages used in the current study and our previous studies was that there were two different messages for each construal condition in the current study (i.e., two messages for the high-construal condition and two messages for the low-construal condition). Therefore, there were four messages in total. Participants were each randomly assigned to one of the four messages (see Figs. 10a and 10c for low-construal messages and Figs. 10b and 10d for high-construal messages). The purpose of this variation in Study 5 was to further establish the effect in novel contexts beyond what has been shown in previous studies. We did not predict that the message factor related to product types would interact with any of the other manipulated factors.

After one of the four messages were shown, we measured the extent of persuasion using three items ("learn more about the product," "try out the recommended product," and "interested in buying the recommended product"; 1 = *not at all*, 7 = *very much*). For the messages regarding both tea tree oil ($\alpha = .94$) and soy milk ($\alpha = .95$), the three items were averaged to create an index of persuasion.

Previous research on persuasion has shown a fit effect, whereby congruence between the characteristics of the message source and the message type could increase "feeling right" about the message and increase persuasion (Lee et al., 2010). Thus, we predicted a fit between the matched construal characteristics of the agent and the construal characteristics of the message ("the message felt right"; "the message felt wrong," reverse-coded; "the message was motivating"; 1 = *not at all*, 7 = *very much*); items were averaged to create the index of fit, $\alpha = .87$.

Results

In our previous studies, we used messages related to sunscreen or exercising in a gym as the products used for persuasion. In Study 5, we randomly assigned each participant to one of the two new product types (i.e., persuasion related to tea tree oil or soy milk) to further generalize our effect to different product types. It was our prediction that there would be no difference in the two product types, and we predicted a three-way interaction between our key independent variables: agent type (i.e., human, AA), agent typicality (i.e., typical, atypical), and message construal (i.e., high construal, low construal). To ensure this, we first conducted a four-way ANOVA using the three key independent

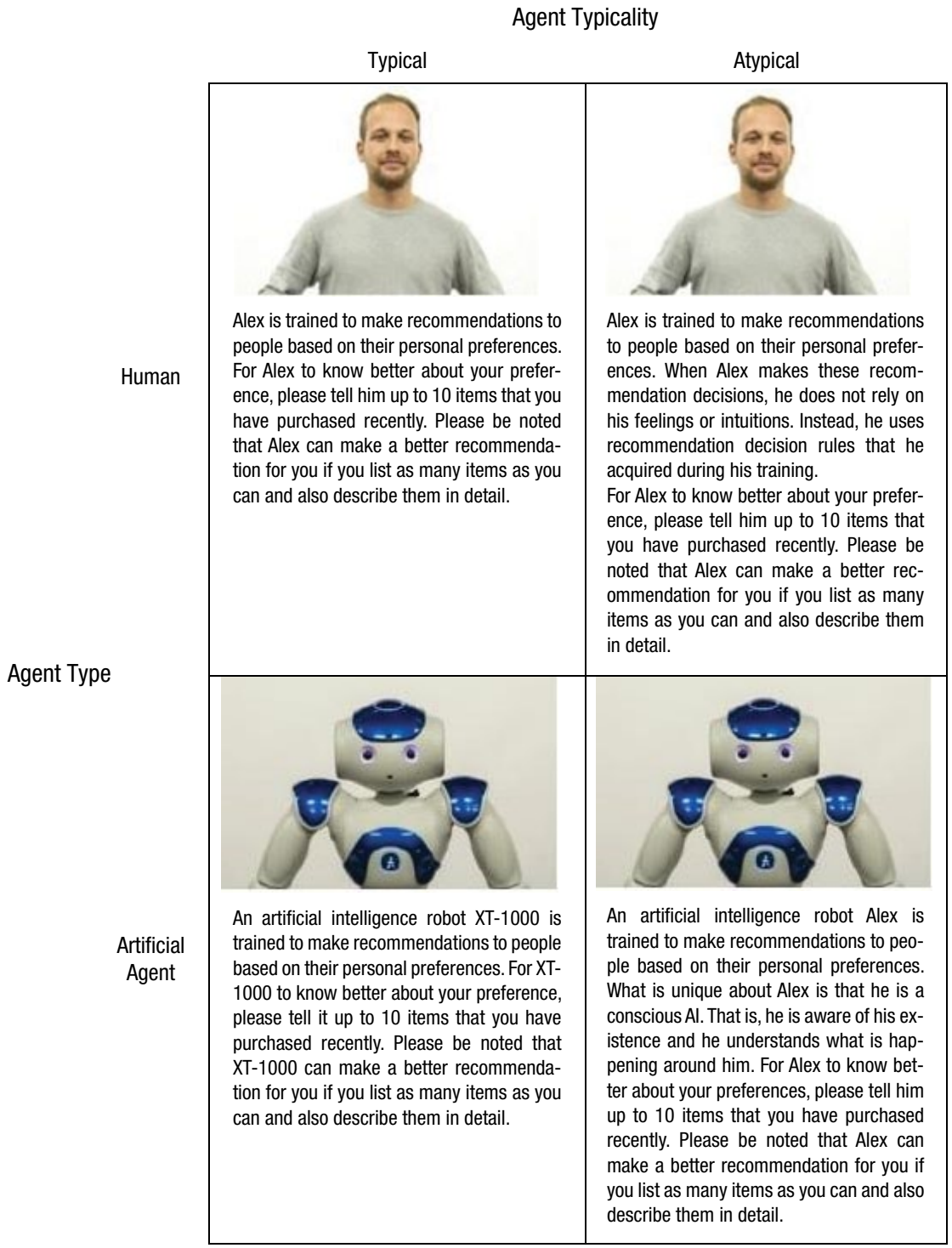


Fig. 9. Schematic showing the manipulation of agent type (human, artificial agent) and agent typicality (typical, atypical) used in Study 5. The typicality manipulation refers to whether the artifical agent was referred to as non-self-aware (typical case) or self-aware (atypical case) and whether the human was referred to as self-aware (typical case) or non-self-aware (atypical case).

variables and the product type (i.e., tea tree oil or soy milk) as the fourth variable. As predicted, the four-way interaction was not significant, $F(1, 384) = 0.06$, $p = .807$, $\eta_p^2 < .001$, but we found a three-way interaction between the three key independent variables, $F(1, 384) = 19.63$, $p < .001$, $\eta_p^2 = .05$. As predicted, the product type did not interact with any of the three key independent variables. Also, we found a main effect of product type;



Fig. 10. Screenshots of the two low-construal messages (a, c) and two high-construal messages (b, d) shown to participants (Study 5).

specifically, the message related to tea tree oil was more effective than the message related to soy milk, $F(1, 384) = 5.82$, $p = .016$, $\eta_p^2 = .02$. This is perhaps because cooking homemade soy milk is more effortful compared with using tea tree oil, leading to a lower level of persuasion overall. To further ensure that random assignment of participants on different product types (i.e., tea tree oil or soy milk) did not influence the interaction between our key independent variables, we conducted a partial F test that compared the two ANOVA models. In one model, we included all three key independent variables (i.e., agent, message construal, and typicality). In another model, we included the same three key independent variables and added a fourth variable, which was product type. The result of this analysis revealed that the two models were not significantly different in explaining the variances of our dependent variable ($p = .186$). Thus, we collapsed the two product types for subsequent analysis. To be more specific, the persuasion index for the tea tree oil and soy milk were averaged to create an index of overall persuasion. This index served as our dependent variable.

Next, we focused on examining the nature of the interaction among our three key independent variables. In this analysis, the overall persuasion index was submitted to a 2 (agent: human, AA) × 2 (typicality: typical, atypical) × 2 (message: high-construal, low-construal) ANOVA. The results revealed only a significant three-way interaction between the three manipulated factors, $F(1, 392) = 19.76$, $p < .001$, $\eta_p^2 = .05$ (see Fig. 11).

To further examine the nature of the three-way interaction, we conducted pairwise comparisons. First, we compared the effectiveness of high-construal and low-construal messages across the four types of agents. This analysis showed a set of results that support our hypothesis. In the typical-AA condition, we replicated the previous findings; specifically, persuasion by the low-construal message ($M = 4.20$, $SD = 1.72$, 95% CI = [3.69, 4.70]) was more effective than persuasion by the high-construal message ($M = 2.95$, $SD = 1.69$, 95% CI = [2.47, 3.42]), $F(1, 392) = 11.27$, $p = .001$, $\eta_p^2 = .12$. In the atypical-AA condition, however, we observed a reversal: The high-construal message ($M = 4.22$, $SD = 1.56$, 95% CI = [3.78, 4.65]) was more effective than the low-construal message ($M = 3.33$, $SD = 1.86$, 95% CI = [2.81, 3.86]), $F(1, 392) = 5.88$, $p = .016$, $\eta_p^2 = .06$.

In the typical-human condition, we replicated previous findings: The high-construal message ($M = 3.67$, $SD = 1.84$, 95% CI = [3.15, 4.18]) and the low-construal message ($M = 3.39$, $SD = 1.92$, 95% CI = [2.85, 3.94]) were not different in their persuasiveness, $F(1, 392) = 0.56$, $p = .46$, $\eta_p^2 = .005$. In the atypical-human

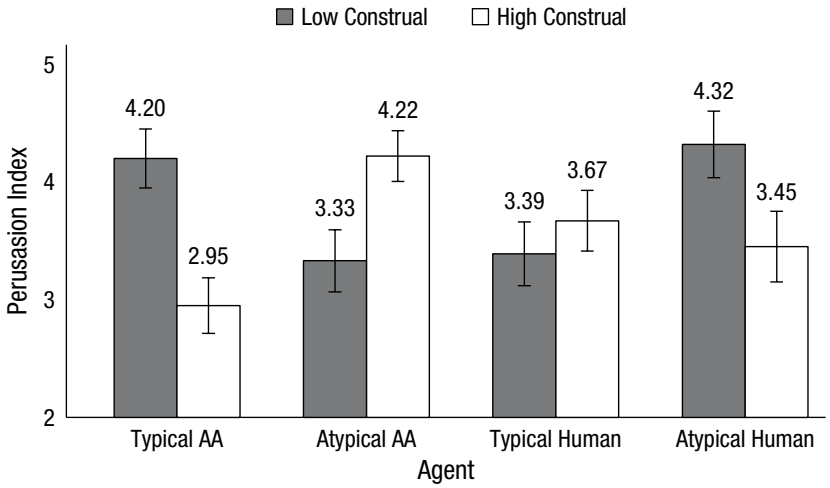


Fig. 11. Mean score on the persuasion index as a function of agent type, typicality, and message's construal level (Study 5). Error bars indicate standard errors. AA = artificial agent.

condition, however, we found that the low-construal message (M = 4.32, SD = 2.02, 95% CI = [3.75, 4.89]) was more effective than the high-construal message (M = 3.45, SD = 2.08, 95% CI = [2.85, 4.06]), $F(1, 392) = 5.50$, $p = .02$, $\eta_p^2 = .04$. These results show that a human agent deprived of human nature can be likened to an AA and an AA ascribed with human nature can be likened to a human, thereby moderating our baseline matching effect (Haslam, 2006).

To examine the underlying process, we submitted the index of fit to the 2 (agent: human, AA) × 2 (typicality: typical, atypical) × 2 (message: high construal, low construal) ANOVA. Just as we found only a significant three-way interaction for the persuasion index (i.e., our dependent variable), we found a significant three-way interaction for the fit index, $F(1, 392) = 12.21$, $p = .001$, $\eta_p^2 = .03$. Also, pairwise comparisons conducted on the fit index revealed that the pattern of results was highly similar to the pattern of the persuasion index (see Fig. 12). Thus, we next conducted a mediation analysis with the fit index as a process variable.

To examine whether the fit index mediates the effect of the key theorized interaction on persuasion, we conducted a mediated moderation test using bootstrapping methods (Muller, Judd, & Yzerbyt, 2005). In this procedure, agent (AA = 0, human = 1), typicality (typical = 0, atypical = 1), and message construal (high construal = 0, low construal = 1) were contrast coded. To test the hypothesized mediation by fit index, we conducted a

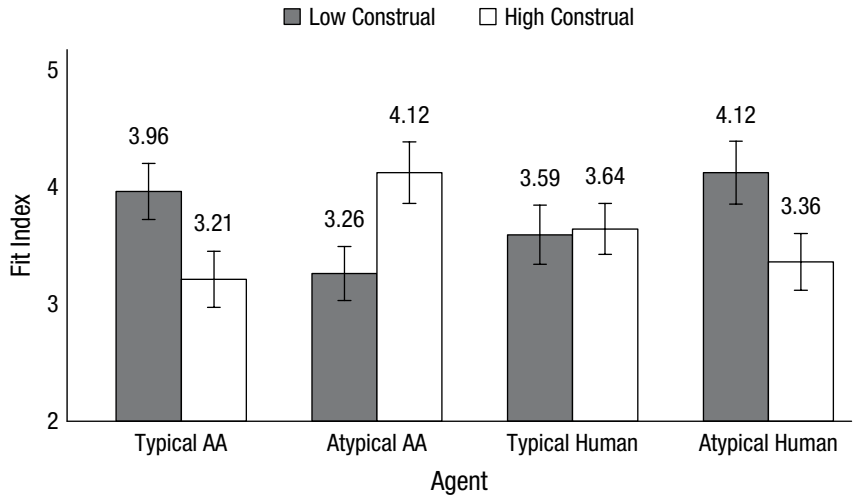


Fig. 12. Mean fit-index score as a function of agent type, typicality, and message's construal level (Study 5). Error bars indicate standard errors. AA = artificial agent.

biased-corrected bootstrapping procedure with 10,000 bootstrap resamples using the Hayes PROCESS macro (Model 4; Preacher & Hayes, 2008). In this analysis, Agent × Typicality × Message Construal was an independent variable, persuasion index was a dependent variable, and fit index was a mediating variable. Also, all independent variables (i.e., agent, typicality, message construal) and all two-way interaction terms between the independent variables (i.e., Agent × Typicality, Agent × Message Construal, Typicality × Message Construal) were entered in the model as covariates. As predicted, the result of this bootstrapping procedure revealed that the 95% CI of the indirect effect (i.e., the path through the mediator) did not include zero, 95% CI = [0.59, 2.23]. Therefore, the mediation by fit index was supported.

Discussion

Study 5 successfully replicated the findings of previous studies using a new AA and new persuasion messages. More importantly, Study 5 showed that an AA could be perceived as having high-construal capabilities when the AA is described with features that are unique to humans. Conversely, a human agent could be likened to an AA when deprived of characteristics that distinguish humans from machines. These results demonstrate that what drives the construal effect are the beliefs people hold about characteristics and the capabilities of the agent rather than the label of the agent (e.g., "a person," "a robot," "an AI") per se.

General Discussion

The current research examined how increasingly pervasive new technologies, such as artificial intelligence and robots, perform as persuasive agents. The research draws on a heretofore unexamined dimension of persuasion by AAs to posit that AAs are perceived as low-construal agents because of the fact that people hold a lay theory that AAs do not have superordinate goals and cannot learn from their experiences or possess consciousness like humans do. As a consequence, individuals perceive greater appropriateness and are more persuaded when an AA's persuasive messages highlight low-construal as opposed to high-construal features. These findings contribute to the literature by identifying key linkages between perceptions of AAs and construal-level theory. The current research (which built off of construal theory) suggests that individuals would rely on AAs more heavily in contexts that induce low construal: events or objects that are temporally, socially, and spatially proximate and occurring with high probability. For example, a short-term investor as opposed to long-term investor may rely on a financial AA more than a human financial advisor (Trope & Liberman, 2010). The current research also contributes to the literature by showing the moderating role of activating distinct lay theories of AAs. The current default lay belief is that AAs are fixed-capability machines that can follow only preprogrammed algorithms. When instilled with the belief that AAs can learn via their experiences or the belief that AAs are conscious about the self and one's actions, we observed an attenuation of the low-construal matching persuasion effect.

Emerging research on AAs has shown that humans often lack trust in AAs because of the fact that they do not have insight into how AAs make decisions (Yeomans, Shah, Mullainathan, & Kleinberg, 2019). Thus, future research can examine whether giving insight into the learning and conscious decision-making mechanisms that AAs use might change human perceptions of AAs as low construal agents. Future research can also examine whether the knowledge about AAs' learning or consciousness can induce fear or feelings of threat in persuasion contexts with AAs. Also, future research can examine the conditions under which AAs are perceived as socially distant agents (e.g., nonhuman out-group member) and as another means of activating high-construal perceptions.

Finally, research has shown a difference between Western and Asian cultures: Asian individuals (e.g., Koreans, Japanese) are more open-minded than Western individuals (e.g., Americans) in embracing AAs (e.g., robots) as social agents (MacDorman, Vasudevan, & Ho, 2009; see also Awad et al., 2018). It is suggested that this cultural difference may stem from fundamental differences in worldviews. For example, a Western monotheistic view supports human dominance over other creatures on the basis of the belief that only humans possess a soul. In contrast, many Asian cultures subscribe to the belief that not only humans but also animals and even objects (e.g., trees) have souls (Kitano, 2007). An interesting avenue for future research would be to examine how different cultural worldviews affect mind-sets toward AAs and drive perceptions of AAs as persuasion agents.

Transparency

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Editor: D. Stephen Lindsay
Author Contributions

T. W. Kim and A. Duhachek both developed and designed the study. Data were collected by T. W. Kim, who analyzed and interpreted the data with input from A. Duhachek. The manuscript was written by both authors, and both approved the final version for submission.

Declaration of Conflicting Interests

The author(s) declared that there were no conflicts of interest with respect to the authorship or the publication of this article.

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Open Practices

All data, experimental materials, and computational-modeling code used to generate the analyses have been made publicly available via the Open Science Framework and can be accessed at https://osf.io/hyn8q/. The design and analysis plans for Study 3 were preregistered on AsPredicted.org at http://aspredicted.org/blind.php?x=pg3wb4. The design and analysis plans for Study 5 were preregistered on the Open Science Framework at https://osf.io/8cdmr/. The complete Open Practices Disclosure for this article can be found at http://journals.sagepub.com/doi/suppl/10.1177/0956797620904985. This article has received the badges for Open Data, Open Materials, and Preregistration. More information about the Open Practices badges can be found at http://www.psychologicalscience.org/publications/badges.



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Supplemental Material

Additional supporting information can be found at http://journals.sagepub.com/doi/suppl/10.1177/0956797620904985

Note

1. Gender and age did not have a main effect or interact with the treatment conditions. These null results related to demographic variables were observed in all studies, and thus these variables will not be discussed further.

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